



**ALLIANCE
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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 5

Title: Building Sales Performance Dashboards

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Experiment No: 5

Title: Building Sales Performance Dashboard

Problem Statement

You are a Sales Analyst for a retail chain. The sales director wants to monitor revenue, average sales, fuel price trends, store performance, weekly sales trends, and holiday sales using a single interactive Power BI dashboard. The objective is to support data-driven decision-making by presenting key business metrics and visual insights in an easy-to-understand format.

Methodology / Steps Followed

Step 1: Data Collection

- Downloaded the Walmart Sales dataset.
- Imported the dataset into Power BI Desktop.
- Verified that all columns were correctly loaded.

Step 2: Data Import

- Opened Power BI Desktop.
- Selected **Get Data** → **CSV**.
- Loaded the Walmart dataset.
- Checked the data preview before loading.

Step 3: Data Cleaning and Transformation (Power Query)

- Checked for missing (Null) values.
- Removed duplicate records.
- Corrected data types.
- Converted the Date column into Date format.
- Verified numerical columns such as Weekly Sales, Fuel Price, Temperature, and Store.
- Loaded the cleaned data into Power BI.

KPI Cards Analysis:



1. Total Sales

- Walmart generated 3.24 Billion in total sales.
- Indicates strong business performance across all stores.
- Helps management monitor overall company revenue.

2. Average Sales

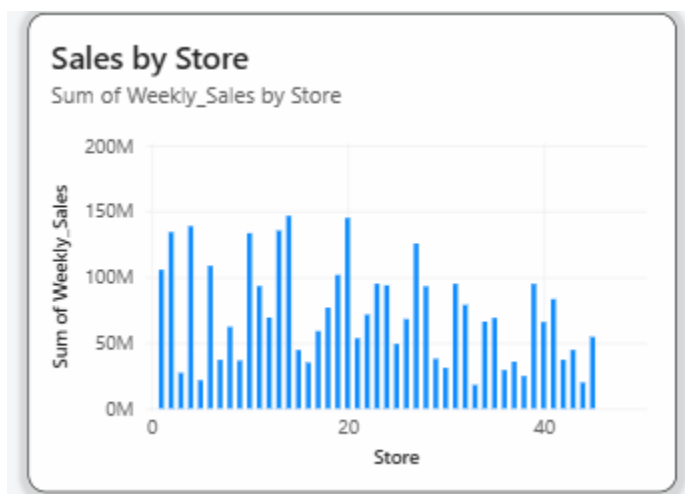
- Average sales are approximately 1.04 Million.
- Useful for comparing future performance with historical averages.

3. Average Fuel Price

- Fuel prices may influence customer travel and shopping behavior.
- Helps analyze the relationship between transportation costs and sales.

Visualizations Analysis:

1. Sales by Store (Column Chart)



Observation

- Sales vary significantly among different stores.
- Some stores generate much higher revenue than others.

Business Insight

- High-performing stores can be used as performance benchmarks.
- Lower-performing stores require improved marketing strategies and inventory planning.

2. Weekly Sales Trend (Line Chart)



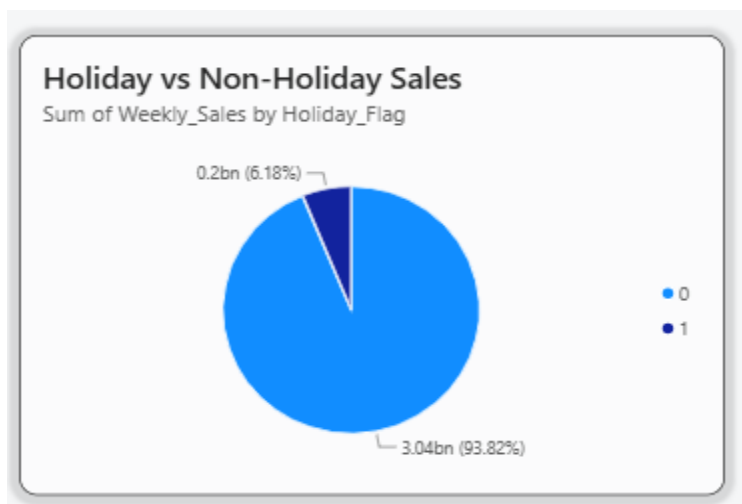
Observation

- Weekly sales remain fairly consistent throughout the year.
- A few weeks show significant sales spikes, especially during peak shopping periods.
- After the peaks, sales return to their normal levels.

Business Insight

- Sales are generally stable.
- Peak periods indicate seasonal demand.
- Management should prepare additional inventory during high-demand weeks.

3. Holiday vs Non-Holiday Sales (Pie Chart)



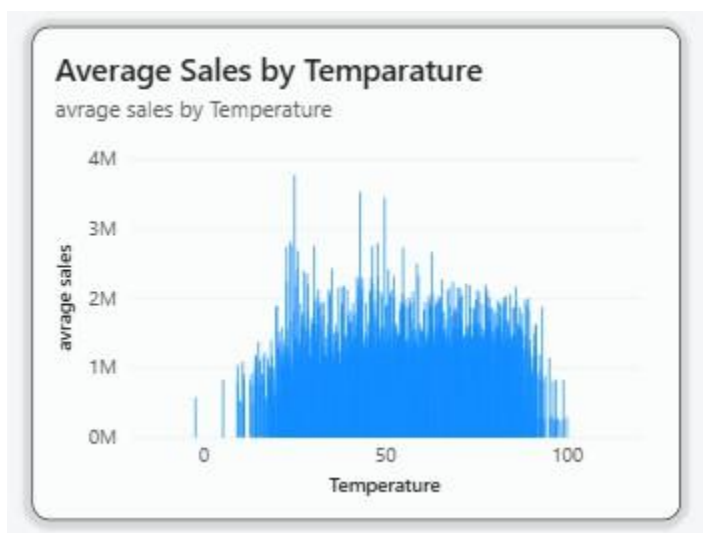
Observation

- Non-Holiday Sales: 93.82%
- Holiday Sales: 6.18%

Business Insight

- Most revenue is generated during non-holiday weeks.
- Holiday sales contribute a relatively small portion.
- Promotional campaigns during holidays could increase holiday revenue.

4. Average Sales by Temperature (Column Chart)



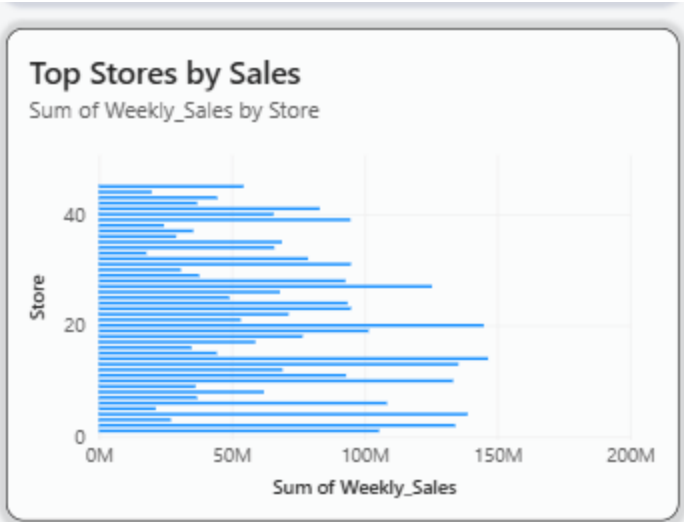
Observation

- Sales fluctuate across different temperature ranges.
- Moderate temperatures generally show stronger sales than extreme temperatures.

Business Insight

- Weather conditions influence customer purchasing behavior.
- Seasonal inventory planning can improve sales performance.

5. Top Stores by Sales (Bar Chart)



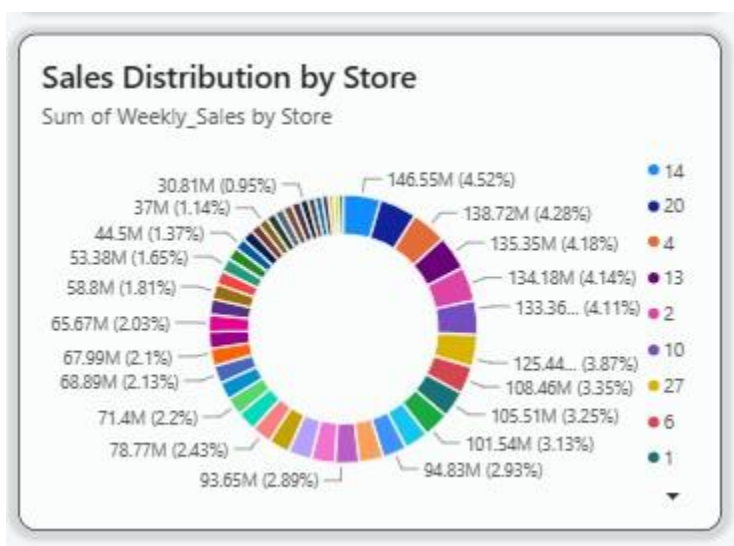
Observation

- A few stores contribute significantly higher sales.
- Remaining stores generate comparatively lower revenue.

Business Insight

- Helps identify the best-performing branches.
- Supports management in recognizing successful stores and sharing best practices.

6. Sales Distribution by Store (Donut Chart)



Observation

- Revenue is distributed across multiple stores.
- No single store dominates total company sales.

Business Insight

- Business performance is balanced across stores.
- Reduces dependence on a single location.
- Indicates a diversified sales network.

Overall Business Insight:



Walmart generated 3.24 Billion in total sales.

- Average weekly sales are approximately 1.04 Million.
- Average fuel price during the selected period is 3.04.
- Non-holiday weeks contribute nearly 94% of total revenue.
- Several stores consistently outperform others.

- Weekly sales remain stable with noticeable seasonal peaks.
- Temperature has an observable impact on customer purchasing behavior.
- Sales are well distributed across stores, indicating balanced business performance.

Conclusion:

The **Walmart Sales Performance Dashboard** successfully provides an interactive view of business performance through KPI cards, column charts, line charts, bar charts, pie charts, and donut charts. The dashboard enables management to monitor overall sales, compare store performance, analyze weekly trends, evaluate holiday sales, and understand the impact of external factors such as fuel prices and temperature. The addition of interactive slicers for Store, Holiday Flag, and Date enhances user experience by allowing detailed analysis from multiple perspectives. Overall, the dashboard supports data-driven decision-making and helps identify opportunities for improving sales performance.

[Github Link](https://github.com/lohithslohiths110405-ui/Data-Analytics-for-Business/blob/main/EXP_5.pbix)

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